
Memorization-Compression Cycles Improve Generalization

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Abstract

We prove theoretically that generalization improves not only through data scaling but also by compressing internal representations. To operationalize this insight, we introduce the Information Bottleneck Language Modeling (IBLM) objective, which reframes language modeling as a constrained optimization problem: minimizing representation entropy subject to optimal prediction performance. Empirically, we observe an emergent memorization–compression cycle during LLM pretraining, evidenced by oscillating positive/negative gradient alignment between cross-entropy and Matrix-Based Entropy (MBE), a measure for representation entropy. This pattern closely mirrors the predictive–compressive trade-off prescribed by IBLM and also parallels the biological alternation between active learning and sleep consolidation. Motivated by this observation, we propose Gated Phase Transition (GAPT), a training algorithm that adaptively switches between memorization and compression phases. When applied to GPT-2 pretraining on FineWeb dataset, GAPT reduces MBE by 50% and improves cross-entropy by 4.8%. GAPT improves OOD generalization by 35% in a pretraining task on arithmetic multiplication. In a setting designed to simulate catastrophic forgetting, GAPT reduces interference by compressing and separating representations, achieving a 97% improvement in separation — paralleling the functional role of sleep consolidation in biological learning process.

1 Introduction

Generalization occurs when learning from one task improves performance on another. The pursuit of generalization in pre-training LLM [11] has historically focused on scaling up data and parameter size [4], in post-training, Reinforcement Learning with Verifiable Reward (RLVR) [7] has gained attention. However, high quality data has been exhausted after years of tireless scraping, and RLVR is shown to only trim knowledge from baseline model [10] instead of ‘incentivize new reasoning pattern’.

We establish a theoretical upper bound on generalization error for deep learning models, indicating representation entropy as another dimension for improving generalization besides scaling data.

Theorem 1. Upper Bound on Generalization Error. *Let $X \in \mathcal{X}$ and $Y \in \mathcal{Y}$ be random variables with an unknown joint distribution $P(X, Y)$, and suppose X is discrete with finite cardinality. Let f be a neural network with L intermediate representations forming a Markov chain:*

$$X \rightarrow R_1 \rightarrow \cdots \rightarrow R_L \rightarrow \hat{Y}$$

Where R_l is internal representations, $\hat{Y}$ is the prediction of the network. Then, for any dataset $\mathcal{D}_N = \{(x_i, y_i)\}_{i=1}^N$ sampled i.i.d. from $P(X, Y)$, the generalization error satisfies for $\alpha \in [1, +\infty)$:

$$\underbrace{\mathcal{L}_{P(X,Y)}(f, \ell)}_{\text{Generalization Error}} \leq \underbrace{\hat{\mathcal{L}}_{P(X,Y)}(f, \ell, \mathcal{D}_N)}_{\text{Empirical Error}} + \underbrace{\mathcal{O}\left(\log N \cdot \min_{1 \leq l \leq L} 2^{\alpha \cdot H(R_l)} / \sqrt{N}\right)}_{\text{Upper Bound on Generalization Gap}} \quad (1)$$

Intuitively, generalization performance of neural network can be improved by either increasing training data size, or by reducing entropy of its internal representations. We refer minimization of $H(R_l)$ as compression, and minimizing empirical loss as memorization.

The Information Bottleneck (IB) framework [8] characterizes the optimal representation as one that discards as much information from the input as possible, while preserving all information relevant to the output. Motivated by this, we introduce the Information Bottleneck Language Modeling (IBLM) objective, along with a theorem showing equivalence of IBLM with IB framework under language modeling.

Definition 1. Information Bottleneck Language Modeling (IBLM). Given a language model with internal representations $R_{1:L}$ and output token variable Y , the IBLM objective is:

$$\begin{aligned} \min \quad & \sum_{l=1}^L H(R_l) \\ \text{s.t.} \quad & \hat{Y} = \arg \min_{\hat{Y}} H(Y|\hat{Y}) \end{aligned} \quad (2)$$

Theorem 2. The IBLM objective defined in Equation 2 is equivalent to the classical Information Bottleneck formulation under the language modeling setting.

Note that $H(Y|\hat{Y})$ is the cross-entropy (CE) loss in conventional language modeling task [11]. In short, IBLM requires maximal compressing internal representation under optimal cross-entropy. To explicitly calculate $H(R)$, we adopt Matrix-based entropy (MBE), first proposed in [5], given a matrix $R \in \mathbb{R}^{s \times d}$ concatenating s token representations, MBE is given by

$$S_\alpha(T) = \frac{1}{1-\alpha} \log \left(\sum_i \left(\frac{\lambda_i(K)}{\text{Tr}(K)} \right)^\alpha \right)$$

where $K = RR^T$ is the Gram matrix. MBE essentially provide a continuous measure of matrix rank, by calculating entropy on distribution of singular values. It has been shown to exhibit a strong correlation with embedding quality [14].

Previous work [15] showed that in neural networks, training often follows a two-phase trend: an initial memorization phase with rapid loss reduction, followed by a monotonic decrease in representation entropy $H(R)$, interpreted as compression. This suggests a clean, sequential separation between fitting and compressing.

Our analysis of GPT pretraining reveals a richer dynamic: the cosine similarity between CE and MBE gradients oscillates between positive and negative, indicating a cyclic alternation between memorization (representation expansion) and compression. Rather than progressing through distinct phases, learning continuously revisits these states. Notably, this local oscillation coexists with a global decline in MBE, suggesting that compression accumulates over time even as the model periodically re-engages in memorization. We term this the memorization–compression cycle.

In biological neural systems, learning proceeds cyclically, alternating between active acquisition and sleep-driven consolidation. A seminal study [18] showed that sleep mitigates interference between conflicting memories by compressing and reorganizing them—allocating distinct subsets of synaptic weights to each. When two conflicting tasks were presented sequentially, awake learning alone led to catastrophic forgetting, while sleep consolidation enabled the formation of separable representations, outperforming even interleaved training. Inspired by this, we propose Gated Phase Transition (GAPT), a training algorithm that alternates between memorization (minimizing cross-entropy) and compression (minimizing cross-entropy and Matrix-Based Entropy). GAPT dynamically switches phases based on learning signals, approximating the IBLM objective and functionally mirroring biological consolidation to resolve representational conflict.

GAPT delivers consistent improvements across domains. In LLM pretraining on the FineWeb dataset, it reduces Matrix-Based Entropy (MBE) by an average of 70.5% across layers while improving cross-entropy by 4.8%, outperforming standard language modeling and closely approximating the IBLM trade-off. In an arithmetic generalization task, GAPT reduces test cross-entropy by 35% and MBE by 47% when trained on 1–3 digit multiplication and evaluated on 4–6 digits, supporting Theorem \ref{thm:entropy-generalization}. Finally, in a synthetic setup with gradient interference, GAPT improves representational separation by 97% and reduces MBE by 91% relative to a mixed baseline—closely mirroring the conflict resolution behavior observed during biological sleep.

Our work makes several key contributions:

- **Theoretical Foundation:** We derive an upper-bound on generalization error showing that reducing representation entropy can improve generalization, alongside scaling data.
- **IBLM Objective:** We formulate Information Bottleneck Language Modeling (IBLM) as a constrained optimization problem, unifying representation entropy and cross entropy targets.
- **GAPT Algorithm:** We propose Gated Phase Transition (GAPT), a algorithm to solve IBLM alternates between memorization and compression phases based on dynamic training signals.
- **Empirical Results:** We show that GAPT improves LLM pre-training, arithmetic generalization, and conflict resolution.
- **Biological Analogy:** We relate the memorization–compression cycle in LLMs to the awake–sleep consolidation cycle in biological systems and validate compression’s similarity to consolidation.

The remainder of the paper expands on these contributions: Section 2 presents our theoretical framework and generalization bound; Section 3 details the GAPT algorithm; Section 4 presents empirical results, including the memorization–compression cycle and GAPT’s effectiveness; Section 5 discusses relevant work.

2 Theory

Corollary 1 (Entropy Lower Bound for Finite Discrete Random Variables). *Let X be a discrete random variable with finite support Ω , where $|\Omega| = n$, and assume that $P(X = x) > 0$ for all $x \in \Omega$. Then there exists a constant $\beta \in (0, 1]$ such that:*

$$H(X) \geq \beta \cdot \log_2 |\Omega|.$$

Proof of Corollary 1 can be found in Appendix A.

Theorem 1. Upper Bound on Generalization Error. *Let $X \in \mathcal{X}$ and $Y \in \mathcal{Y}$ be random variables with unknown joint distribution $P(X, Y)$, and suppose X is discrete with finite cardinality. Let f be a neural network with L intermediate representations forming a Markov chain:*

$$X \rightarrow R_1 \rightarrow \dots \rightarrow R_L \rightarrow \hat{Y}$$

Then, for any dataset $\mathcal{D}_N = \{(x_i, y_i)\}_{i=1}^N$, there exists $\alpha \in [1, +\infty)$ s.t. the generalization error satisfies

$$\mathcal{L}_{P(X,Y)}(f, \ell) \leq \hat{\mathcal{L}}_{P(X,Y)}(f, \ell, \mathcal{D}_N) + \mathcal{O}\left(\frac{\log N \cdot \min_{1 \leq l \leq L} 2^{\alpha \cdot H(R_l)}}{\sqrt{N}}\right)$$

Proof. We begin by recalling the standard formulation of the generalization gap:

$$\text{Gen}_{P(X,Y)}(f, \ell, \mathcal{D}_N) := \mathcal{L}_{P(X,Y)}(f, \ell) - \hat{\mathcal{L}}_{P(X,Y)}(f, \ell, \mathcal{D}_N)$$

For discrete input variables X , it has been shown in [6] that the generalization gap is upper-bounded by:

$$\begin{aligned} \mathcal{L}_{P(X,Y)}(f, \ell) - \hat{\mathcal{L}}_{P(X,Y)}(f, \ell, \mathcal{D}_N) &\leq \mathcal{O}\left(\frac{|\mathcal{X}| \log N}{\sqrt{N}}\right) \\ &\leq \mathcal{O}\left(\frac{2^{\alpha \cdot H(X)} \log N}{\sqrt{N}}\right) \end{aligned}$$